



Waveform Parameters

Icon	Name	Timbre	Harmonics	Morph	Out 2
 1	Virtual analog	Square wave: narrow pulse, full square, hardsync formant	Detuning between waves	Saw: triangle to wide notch saw	Sum of two hardsync'd waveforms
 2	Wave-shaping	Wavefolder amount	Waveshaper waveform	Waveform symmetry	Variant with another waveform curve
 3	FM 2 Operators	Modulation mix	Frequency ratio	>12:00 – operator 2 modulates own phase<12:00 – operator 2 modulates operator 1 phase	Sub-Oscillator
 4	Grains	Formant frequency	Frequency between formant 1 and 2	Formant width and shape	Simulation of filtered waveforms – Harmonics selects filter type (peaking, LP, BP, HP)
 5	Additive	Most prominent harmonic	Number of bumps in spectrum	Bump shape – flat and wide to peaked and narrow	Variant that includes harmonics from Hammond organ drawbars
 6	Chords	Chord inversion /transposition	Chord type	Waveform	Chord root note
 7	Speech	Vocal tamper from deep to high	Scrolls through formant types, SAM, and LPC vowels/words	Word segment selection	Unfiltered vocal signal
 8	Karplus Strong	Brightness and dust noise sensitivity	String stiffness	Decay time	Copy of Out 1
 9	Super Saw	Sets number of waveforms	Adjusts harmonic content	Sub-oscillator level	Copy of Out 1
 10	Wavetable Oscillator	Rotates through different waves	Selects between 4 interpolated banks followed by the same 4 banks, in reverse order, without interpolation.	Column index	Bit reduced version of Out 1

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 1	Rain	Rain grain density	Amount of pitch randomization	Droplet duration and overlap, culminating in a stack of 8 randomly frequency-modulated waveforms	Varient with sine wave oscillators
 2	Noise	Clock frequency	Scrolls through filter response, from LP to BP to HP	Filter resonance	Result of 2 BP filters controlled by Harmonics knob
 3	Dust	Particle density	Frequency randomization	Scrolls through reverberating all-pass filters followed by increasingly resonant BP filters	Raw dust noise
 4	Modal Strings	Excitation brightness and dust density	Amount of harmonic coloration	Decay time	Raw exciter signal
 5	FM Drum	LP filter cutoff	Blend between harmonic content	Decay time	Alternate FM Drum model
 6	Bass Drum	Attack brightness and overdrive amount	Frequency	Decay time	Alternate Bass Drum model
 7	Snare Drum	Balance between different modes of the drum	Blend between harmonic and noisy components	Decay time	Alternate Snare Drum model
 8	Hi-hat	HP filter cutoff	Blend between metallic and filtered noise	Decay time	Alternate Hi-hat model
 9	Cowbell	Brightness	Texture	Decay time	Alternate Cowbell model
 10	Toms	Tone	Resonance	Decay time	Alternate Tom models

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1	BX7	Vibrato	Preset selection	Tremolo	Velocity
2	Bassline	Cutoff	Resonance + distortion	Env mod + Decay	Accent
3	Wave Generator	Waveform	Bit Crush	Sample Rate	Level
4	Vox	Formant shift	Reso	Blend Vowels	Level